The EIC and ePIC for a polarized-lithium programme: beam and detector parameters, their provenance, and the lithium case derived

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2026-08-27 · rewritten 2026-08-28 with the far-forward optics re-derived per configuration · the specification behind Reports 1, 2 and 4 · every entry carries its source and its **status** — design value, requirement, measured, derived here, or unknown · figures from `evgen/scripts/eic_beam_figures.py`, tables from `polli_fastsim.beams`, `polli_fastsim.farforward` and `fastsim/scripts/tagging_acceptance.py`

Abstract. Every projection of the polarized-lithium programme rests on parameters that belong to the accelerator or to the detector. This report collects them, states what each one is — a design value, a requirement, a measurement, a derivation of ours, or an unknown — and derives the lithium case where nothing is published. Three results carry the rest of the programme. Ion energies are set by the revolution period, not by rigidity: ions are γ-matched to the proton configurations, so the ^{6}Li menu is 41 and 99–138 GeV/u with nothing between. The beam divergence, which every far-forward acceptance is an exponential in the square of, follows from the Yellow Report's proton tables with a species factor that is unity wherever the ion is γ-matched and √2 only at the rigidity-capped top: 220/380, 180/180 and 92/92 μrad (h/v) for ^{6}Li at the three configurations. Under those optics the intact-^{6}Li recoil cannot be tagged at any configuration (7×10^{-8} to 6×10^{-27} with the measured pot aperture) and the ^{6}Li α spectator only through its off-rigidity window (1.5–1.7%); a lithium tagging optics — the horizontal β* raised 50–176× with the pots following the envelope, at 1/7–1/13 of the luminosity — recovers a 32–42% coherent tag and a 27–35% α tag, and is the one *new* optics this programme asks for; ^{7}Li keeps the published one, for which it pays nothing, so the ask is two run configurations rather than two lattice designs. The detector model is a requirement where a requirement exists and a labelled stand-in where none does: the track angular resolution and the electron-identification curve have no published ePIC source.

1 · Scope and status vocabulary

Report 1 presents the measurement and Report 2 the simulation and reconstruction chain; this report is the specification both are conditional on, in a form that separates what the machine and detector documents state from what this programme has had to derive or assume. Five status words are used throughout: **design value**, a parameter that appears in a design document; **requirement**, a specification a subsystem must meet, not a demonstrated performance; **measured**, from test beam, operations, or — for the pot aperture — a simulation of the detector geometry; **derived**, computed here from published inputs with the derivation shown; and **unknown**, not published anywhere we could reach. §2 gives the collider and the ion energy menu, §3 the beams, §4 the lithium case, §5 the far-forward acceptance of the lithium tags at each optics, §6 the ePIC detector, and §7 the list of what the programme assumes on the machine's and the detector's behalf, with owners.

2 · The collider and the ion energy menu

The EIC is two rings in the RHIC tunnel: the Electron Storage Ring at 5, 10 and 18 GeV and the Hadron Storage Ring carrying protons to 275 GeV and ions to the same magnetic rigidity, 917.3 T·m. ePIC sits at IP6; a second interaction region at IP8 is designed but not funded. The crossing angle is 25 mrad at IP6 and 35 mrad at IP8, recovered by crab cavities [1,2].

Colliding bunches requires equal revolution periods. The electrons are ultrarelativistic at every energy, so their period is fixed; the hadrons must match it, and since the circumference is adjustable only within a narrow range, the hadron velocity — hence γ — is quantised by the ring geometry, and the magnets supply whatever rigidity that γ implies for the species in the ring, up to the cap. Two limits therefore bind at the two ends of the menu: at the top the magnets bind and the ion is rigidity-capped at p/nucleon = 275 GeV × Z/A; at the bottom the circumference binds and the ion is γ-matched to the proton configuration, at a per-nucleon energy that rigidity scaling does not give. EPIOS describes the mechanism [2]: a radial shift of up to ±20 mm in the arcs covers 118 < γ < 293, and a bypass arc 90 cm shorter reaches γ = 43.5. The check that settles the rule is Yellow Report Table 10.2, which lists gold at 41 GeV/u — γ = 44.02 against the 41 GeV proton's 43.70, equal to 0.7%, the u-versus-m_p mass difference — where rigidity scaling would give 16.4 GeV/u; the same rule reproduces the published ^{3}He menu, "41, and 100–183 GeV/nucleon", exactly. One conflict is recorded rather than resolved: Yellow Report Table 10.1 runs 100 GeV protons, γ = 106.6, in neither stated window. The programme stays anchored on the Yellow Report's three configurations because every beam-parameter table is indexed by them, and `tools/consistency_check.py` keeps the point visible.

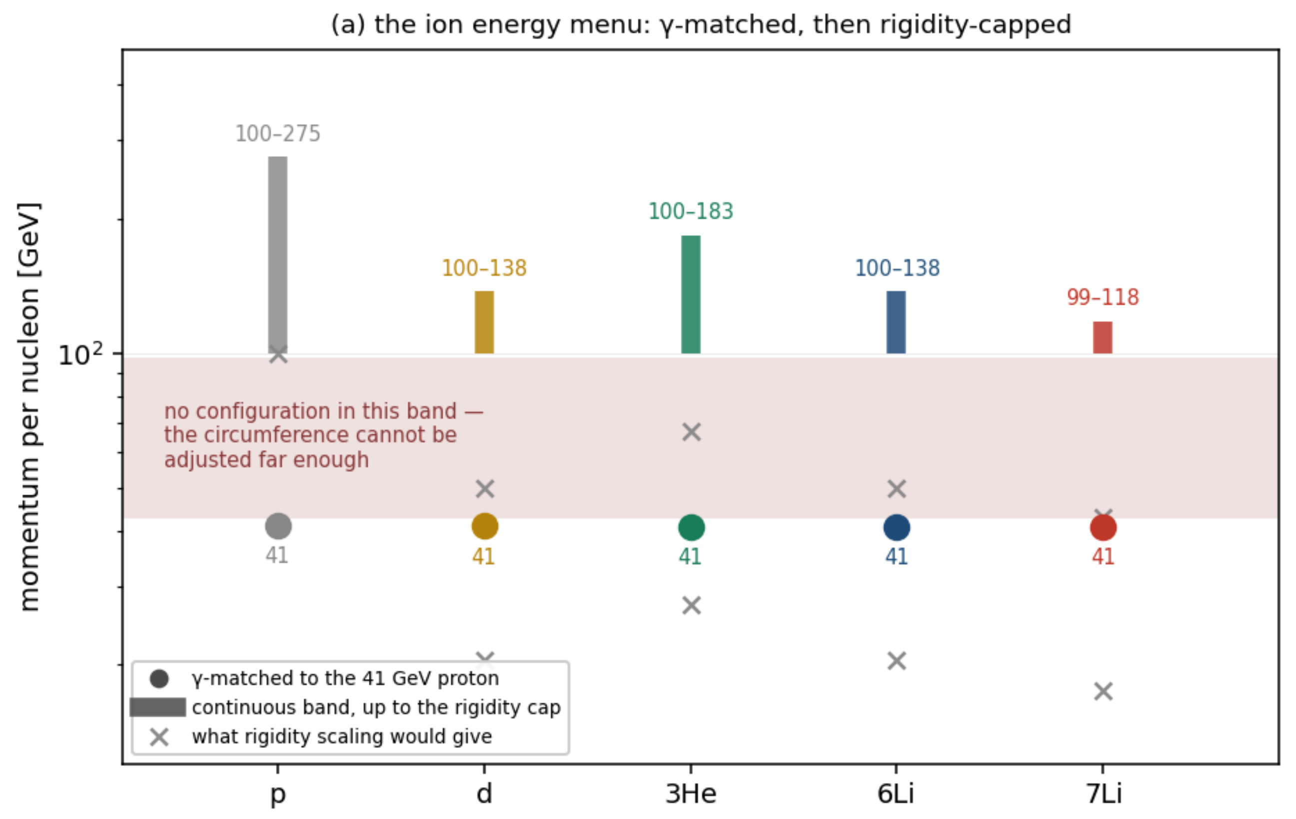

Figure 1 — the ion energy menu. Each species has an isolated γ-matched point at ≈ 41 GeV/u and a continuous band from ≈ 99 GeV/u to its own rigidity cap, with nothing between. Crosses mark what rigidity scaling would have given, which this programme used until 2026-08-27. `eic_beam_figures.py`.

SPECIES	A/Z	Γ-MATCHED POINT	CONTINUOUS BAND	RIGIDITY CAP	STATUS
p	1	41 GeV	100–275 GeV	275 GeV	design value
d	2	41.0 GeV/u	100–137.5	137.5	design value
^{3}He	3/2	40.9	99.8–183.3	183.3	design value (menu published)
^{6}Li	2	**40.8**	**99.5–137.5**	137.5	**derived**; top ≈ 138 in EPIOS
^{7}Li	7/3	**40.8**	**99.5–117.9**	117.9	**derived**; top ≈ 117 in EPIOS
Au	2.49	41	– (110 published)	110.3	design value

Table 1 — the energy menu. `beams.default_configs` computes these; run on a proton it returns 41 / 100 / 275 exactly, the implementation's self-check. Only the lithium rows are derived, and only their top energies appear in EPIOS [2].

3 · The beams

3.1 · Electron beam

QUANTITY	VALUE	SOURCE	STATUS
energies	5, 10, 18 GeV	YR §10.1	design value
β* (h/v)	45 / 5.6 cm	EPIOS Table I (EIC column)	design value
RMS emittance (h/v)	20 / 1.3 nm	EPIOS Table I	design value
bunch charge, bunches	28 nC, 1160	EPIOS Table I	design value
divergence h/v, 18 GeV	202 / 187 (HD), 89 / 82 µrad (HA)	YR Table 10.1	design value
Δp/p	$5.8–10.9 \times 10^{-4}$	YR Table 10.1	design value
polarization	70%	YR physics studies	assumption for projections; unused by the tensor observable

Table 2 — the electron beam. The electron divergence enters this programme only through the track angular resolution of Report 2 §3, for which it is the irreducible floor.

3.2 · Hadron beam

Yellow Report Tables 10.1 (e+p) and 10.2 (e+Au) are the authoritative divergence and momentum-spread tables and carry both optics settings; high divergence and high acceptance differ by a β* choice, and the Yellow Report states that $\sigma_\theta \propto 1/\sqrt{\beta^*}$ while the luminosity goes as 1/β*, so acceptance is bought with luminosity [1]. The baseline cooling is conventional electron cooling at injection, demonstrated at LEReC; EPIOS is explicit that store cooling is a later-stage addition worth a factor two in average luminosity [2], so the emittance grows under intrabeam scattering through the fill with nothing opposing it. The other design values (EPIOS Table I, EIC column): β* = 80 / 7.2 cm, RMS emittance 11 / 1 nm, bunch charge 11.0 nC, 1160 bunches, bunch length 6 cm, peak luminosity 10^{34} cm^{-2} s^{-1}. The beam is flat, $\varepsilon_x/\varepsilon_y \approx 11$, with β* matched so that the two divergences come out nearly equal — which is why Table 3's h and v agree at 275 and 100 GeV but not at 41.

PROTON CONFIGURATION	HD H/V [MRAD]	HA H/V	ΔP/P [10^{-4}]	L [10^{33} CM^{-2}S^{-1}] HD / HA
275 GeV, 290 bunches	150 / 150	65 / 65	6.8	1.54 / 0.32
275 GeV, 1160 bunches	119 / 119	65 / 65	6.8	10.0 / 3.14
100 GeV	220 / 220	180 / 180	9.7	4.48 / 3.14
100 GeV (vs e 5)	206 / 206	180 / 180	9.7	3.68 / 2.92
41 GeV	220 / 380	220 / 380	10.3	0.44 / 0.44

Table 3 — YR Table 10.1, hadron rows, verbatim (design values). At 41 GeV the two optics are identical: the only configuration with no separate high-acceptance option, and the one the coherent programme would most like one at (§5).

GOLD CONFIGURATION	STRONG HADRON COOLING H/V	ΔP/P	STOCHASTIC COOLING H/V	ΔP/P
110 GeV/u	218 / 379 µrad	6.2×10^{-4}	77 / 380	10×10^{-4}
41 GeV/u	275 / 377	10×10^{-4}	174 / 302	13×10^{-4}

Table 4 — YR Table 10.2, gold rows. The two cooling scenarios differ by 2.8× horizontally at 110 GeV/u: the cooling choice is worth as much as the optics choice.

4 · Lithium: published, derived, unknown

What is published is thin. EPIOS [2] gives the spin factors and the top energies and states the case; it gives no lithium luminosity, emittance or intensity, and describes the polarized-lithium source as early-stage. Hamwi and Hoffstaetter [3] give the depolarizing-resonance spectrum and the statement that small-G ions cannot use Siberian snakes.

QUANTITY	^{6}LI	^{7}LI	SOURCE / STATUS
spin, A/Z	1, 2	3/2, 7/3	nuclear data
G factor	−0.178 (EPIOS), −0.1818 (Hamwi)	+1.532 / +1.5196	published; the two disagree in the last digits
max |Gγ|, resonances to cross	27, 81	191, 573	Hamwi Table I, design study
snake scheme	^{6}Li deuteron-like — not amenable to Siberian snakes; partial solenoid and tune jumps; ^{7}Li partial snakes		Hamwi; EPIOS
energy menu	41, and 99–138 GeV/u	41, and 99–118	derived (§2)
σ_θ h/v, high acceptance	220/380, 180/180, 92/92 μrad	220/380, 180/180, 99/99	derived (§4.1)
Δp/p	$7–10 \times 10^{-4}$	same	derived; species-insensitive (RF bucket)
tagging optics (§4.2)	β^*_x × 50 / 176 / 89; envelope 0.33 × 3.8, 0.17 × 1.8, 0.12 × 0.92 mrad; L/L_{HA} = 1/7.1, 1/13.3, 1/9.5	similar	this programme's proposal
emittance, intensity, β*, luminosity	— nothing published —		unknown (§7)

Table 5 — the lithium beam.

4.1 · The divergence, derived

$\sigma_\theta = \sqrt{\varepsilon_N/(\beta\gamma\,\beta^*)}$. At a given machine configuration β* is common to every species, the lattice being set by rigidity, so the only species dependence is through βγ; and a γ-matched ion has the proton's βγ exactly (§2). The species factor on the proton divergence is therefore

$$\frac{\sigma_\theta^{\mathrm{ion}}}{\sigma_\theta^{p}} = \sqrt{\frac{(\beta\gamma)_p}{(\beta\gamma)_{\mathrm{ion}}}} = 1 \;\; (\gamma\text{–matched}:\; 5\times41,\; 10\times100), \qquad \sqrt{2}\;\; (\text{rigidity capped},\; A/Z=2:\; 18\times275)$$

— not a blanket factor, which an earlier version of this programme's correction assumed. Whether equal ε_N is defensible for lithium is tested on gold, the published case: scaling the 275 GeV proton to Au at 110 GeV/u by $1/\sqrt{(\beta\gamma)}$ alone predicts 236 μrad against the observed 218 (h) and 379 (v), so $\varepsilon_N(\mathrm{Au})/\varepsilon_N(p)$ = 0.85 horizontally and 2.6 vertically. Intrabeam scattering at fixed beam current goes as Z^3/A^2: 0.75 for ^{6}Li, 1 for a proton and 12.7 for gold — 17× lithium's — and RHIC deuterons showed no measurable IBS growth while gold grew 20–45% per hour. Lithium is a proton-class ion; equal ε_N is the right starting assumption, and it is an assumption. Δp/p is the easy one: protons and gold give $6.2–13 \times 10^{-4}$ across every energy and cooling scheme, set by the RF bucket rather than the species.

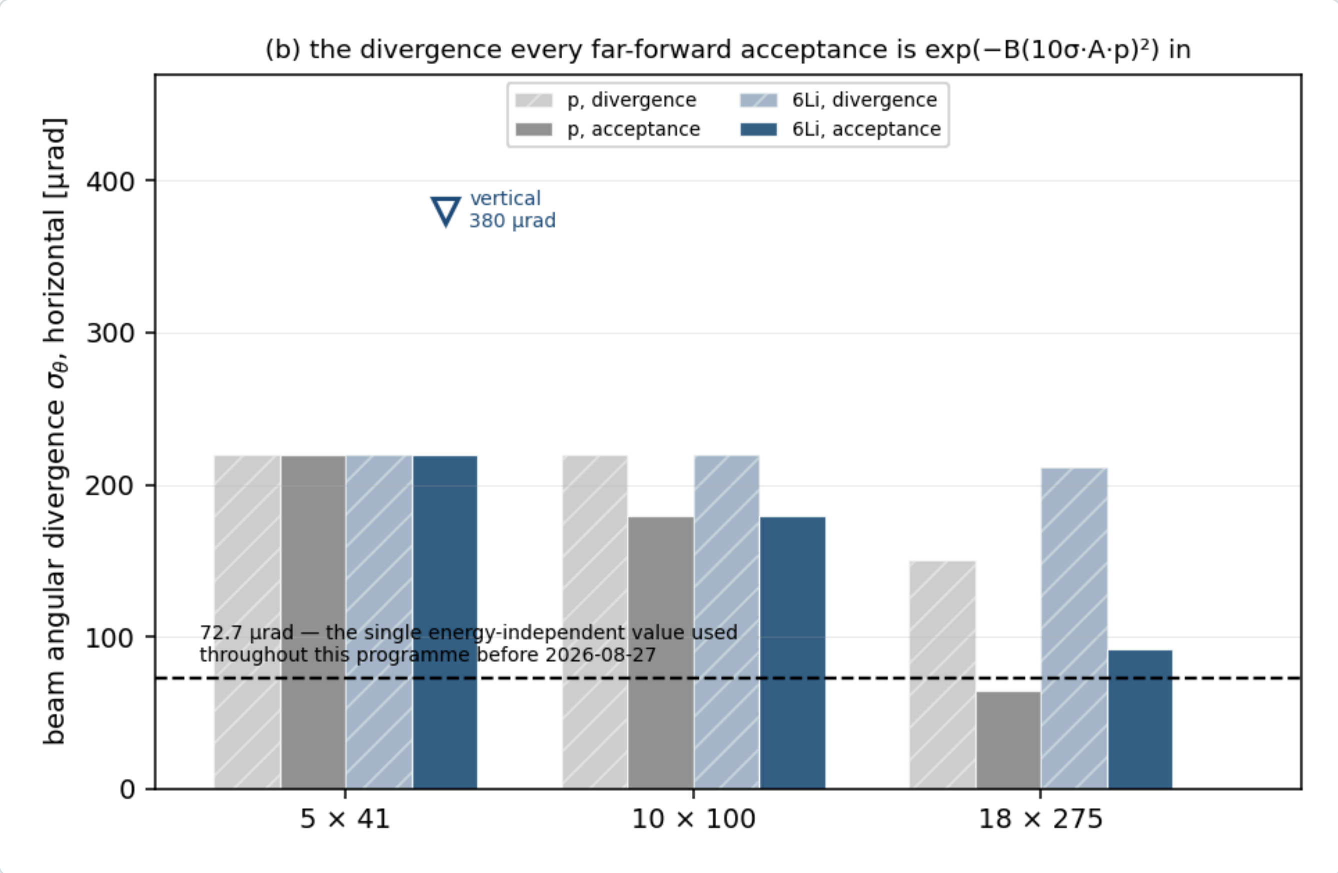

Figure 2 — the divergence. Protons and ⁶Li per configuration and optics; the vertical value is marked where it differs from the horizontal. The species √2 appears only at 18 × 275; the dashed line is the single energy-independent 72.7 μrad used before 2026-08-27, below almost everything. `eic_beam_figures.py`, `farforward.sigma_theta_for`.

4.2 · A lithium tagging optics

The far-forward lithium tags — the intact ^{6}Li recoil of the coherent channel and the α spectator of the tagged channels — are beam-blind (A/Z = 2 exactly; the α at 0.99813 of the beam rigidity) and are separated from the beam only by their angle at the interaction point against the 10σ envelope and the pot aperture. At the optics of Table 3 neither survives (§5). The one lever is β*: because a recoil escapes through the horizontal gap between the pots or through the vertical one, only the plane it escapes through needs a larger β*, and Report 1 §6.1 prices the horizontal de-squeeze alone — $\sigma_{\theta,x} \propto 1/\sqrt{r}$ at a luminosity $1/\sqrt{r}$, the vertical plane at high acceptance, the dispersion of the beam's momentum spread at the pots (D ≈ 0.3 m against R_{12} = 30.6 m) added to the horizontal envelope, and the pots following the envelope in both planes. The optimum of tagged fraction × luminosity is at r = 49.7, 175.6 and 89.3 for the three ^{6}Li configurations, where the envelope is 0.33 × 3.8, 0.17 × 1.8 and 0.12 × 0.92 mrad, the tagged fraction 0.42, 0.32 and 0.33, and the luminosity 1/7.1, 1/13.3 and 1/9.5 of the high-acceptance value (`farforward.tagging_optics_point`). Two assumptions carry it: the electron β* raised in step to keep the spots matched, and a parallel-to-point far-forward transport for the de-squeezed lattice ($R_{11}\sigma^* \ll R_{12}\sigma_\theta$ at the pots, as in the LHC's high-β* optics) — true of the present lattice at the Yellow Report optics and the first thing to establish for a de-squeezed one. β^*_x ≈ 45 m at 5 × 41 (0.90 m × 49.7; ≈ 107 m and ≈ 71 m at the other two configurations) is the smallest absolute ask and sits inside the LHC's demonstrated forward-physics range (β* = 90 m and 2500 m against a nominal 0.55 m). Unlike the isotropic ×14.5 optimum of plans/10 (β* = 13 m), which would have shrunk the beam in the final-focus quadrupoles ($\beta \approx \beta^* + L^2/\beta^*$ ≈ 15 m against 29 m today for L ≈ 5 m), a 45 m de-squeeze leaves the horizontal β at the first quadrupole at ≈ 46 m, larger than today, so the IR aperture joins matching and chromaticity among the questions for C-AD. It is a proposal, not a specification, and the request is stated in §7.

5 · The far-forward lithium tags at each optics

Table 6 evaluates both tags at the Yellow Report optics of each configuration and at the tagging optics of §4.2 with the pots following the envelope. The coherent tag is the analytic acceptance of an exp(−B|t|) recoil, B = 50 GeV⁻², outside the rectangular cutout, with the pot aperture measured in the ePIC geometry (§6.1) as a second constraint per axis at the Yellow Report optics (cutouts 2.20 × 3.8, 1.80 × 3.0 and 1.03 × 2.3 mrad); the spectator tags are evaluated against the envelope alone, which binds or nearly binds at every configuration, and are the cluster model of `polli_fastsim.spectator` (β = 0.30 GeV, the short-range parameter of the two-parameter cluster momentum density, whose 0.20–0.40 band is the model uncertainty [8]) routed through the far-forward windows of `polli_fastsim.farforward` — Roman Pots at 0.60 ≤ R ≤ 0.95, off-momentum detectors at 0.45 ≤ R < 0.65, B0 at 5.5–20 mrad, the near-beam band |R − 1| < 0.05 tagged only outside the envelope — with the rectangular envelope applied to each fragment's azimuth.

TAG	OPTICS	5 × 40.8	10 × 99.5	18 × 137.5 (^{6}LI) / 18 × 117.9 (^{7}LI)
coherent ^{6}Li recoil, tagged fraction	YR high acceptance + measured aperture	7.2×10^{-8}	6.2×10^{-27}	1.9×10^{-17}
	tagging optics, pots follow (L/L_{HA})	0.42 (1/7.1)	0.32 (1/13.3)	0.33 (1/9.5)
^{6}Li α spectator (d struck), tagged	YR high acceptance	0.017	0.015	0.016
	YR high divergence	0.017	0.015	0.015
	tagging optics	0.35	0.27	0.28
^{6}Li d spectator (α struck), tagged	YR high acceptance	0.078	0.061	0.068
	tagging optics	0.55	0.49	0.50
^{7}Li α spectator (t struck), tagged	YR high acceptance	0.967	0.966	0.971
	tagging optics	0.986	0.990	0.991
^{7}Li t spectator (α struck), tagged	YR high acceptance	0.033	0.004	0.005

Table 6 — the far-forward lithium tags per configuration and optics (`tagging_acceptance.py`, `nearbeam_aperture_scan.py`; 2026-08-28). At the Yellow Report optics the ^{6}Li α tag is the ≈ 1.5% that falls below R = 0.95 into the Roman-Pot window — the near-beam tail is below the envelope at every configuration — and the ^{6}Li d tag — the deuteron shares the beam's A/Z = 2, so it too sits at R ≈ 1 — is the ≈ 6% that falls below R = 0.95 into the same window, with nothing in the off-momentum detectors; the ^{7}Li α at R = 0.856 lands mid-window regardless of optics, and the triton (R = 1.29) is over-rigid, with no window in `farforward.route_charged`, which carries no R > 1 branch — so its row is the low-momentum tail of the cluster distribution dipping into the Roman-Pot window or the near-beam band, with the B0 supplying most of it at 5 × 40.8. Its "no coverage" is a routing assumption rather than a measurement: the 2026-06-12 particle-gun scan of `tools/fullsim` puts an over-rigid triton on the inner side of the Roman-Pot planes and then in the zero-degree calorimeter, a path nobody has yet checked with a beam envelope, and one that would make the ^{7}Li α + t double tag possible (plans/03 §2.2 [8]). The third column is the top configuration of each isotope, 137.5 GeV/u for ^{6}Li and the 117.9 GeV/u rigidity cap for ^{7}Li. Before 2026-08-28 the fast simulation applied a single proton-derived 73 μrad at every configuration, and applied it azimuth-blind as a circle: that pair of choices gives 1.9% for the ^{6}Li α at 18 × 137.5 (published as 1.85%) and 20% at 5 × 40.8, against 1.6% and 1.7% here. Two readings of the ^{6}Li α number coexist in this repository and neither stands in for the other. The rows above are `polli_fastsim.spectator`, one partial wave per channel — L = 0 for α–d — while the tagged generator behind Report 4 §2.1 (`polligen.tagged`) carries the full S + D expansion of the same channel: the S-wave radial is identical in the two, ⟨k⟩ = 0.1071 GeV/c, and the D wave is hard at ⟨k⟩ = 0.2778, so at P_D = 0.0867 the channel mean moves to 0.1219 and the off-rigidity slice from 1.5% to 2.5%. The D wave is not an optional tail but the tensor observable itself — set P_D = 0 and the α–d density is m-independent and A_{zz}^{tag} vanishes identically — so the tagged asymmetries are quoted on the S + D spectrum and this table on the S-wave one: 1.5–1.7% here, 2.5–2.9% there. The two ^{7}Li densities are the same pure P wave, and the 0.6 points by which this table's 0.9674 at 5 × 40.8 exceeds the tagged script's 0.9614 is a difference of *acceptance definition*: the column above is 1 − lost, i.e. any far-forward system, while `tagged_polarimetry_7li.py` masks on the Roman Pots alone, and at 5 × 40.8 the B0 carries 1.1% of the ^{7}Li α. Appendix A carries the like-for-like comparison.

Two conclusions. The tagging inversion between isotopes at IP6 is optics-independent: the ^{7}Li α works at 97% at every optics, the ^{6}Li α does not work at any published one. And the tagging optics converts the ^{6}Li tags from beam-blind to usable — 32–42% of coherent recoils, 27–35% of α spectators, 49–55% of deuteron spectators — at a luminosity cost of 7–13, which is the trade Report 1 §6.1 prices and Report 4 examines from the detector side. One caution on reading the two ^{6}Li spectator rows: they are not independent channels but the two fragments of the same breakup, and sampled jointly (Report 4 §4.2) both are recorded in 22–29% of breakups at the tagging optics — so four fifths of the α row there lies inside the deuteron row, while about half the deuteron row is deuteron-only, the deuteron being the systematically wider fragment; at the Yellow Report optics, where the joint fraction is below 0.03%, the two rows are effectively disjoint.

The luminosity cost is not shared evenly between the isotopes, and for ^{7}Li it is not paid for at all. The ^{7}Li α row moves from 0.967–0.973 to 0.986–0.991 under the tagging optics — ×1.02 — for a luminosity of 1/8.2, 1/15.4 and 1/10.1, so the de-squeeze is a factor 8–15 *net loss* in tagged ^{7}Li events and multiplies every ^{7}Li error bar by 2.8–3.9 at equal running time (Report 4 §2.1). The α tag that lands mid-window is the one the machine cannot improve. So the inversion in the first conclusion runs in both directions: ^{6}Li needs the de-squeeze and ^{7}Li must not have it, which makes the two isotopes separate runs with separate optics rather than one fill plan — a scheduling consequence to put to C-AD alongside the lattice questions of §7.

6 · The ePIC detector

6.1 · Far forward

SUBSYSTEM	POSITION	ACCEPTANCE	TECHNOLOGY	STATUS
B0	`z ≈ 5.4–6.4 m`	`5.5 < θ < 20 mrad, any charged; γ`	AC-LGAD + EMCal	design value
off-momentum detectors	`z = 25.5, 27.0 m`	`θ < 5 mrad, 0.45 ≤ R < 0.65`	AC-LGAD	design value (windows [6]); positions from the geometry file
Roman Pots	`z = 32.55, 34.25 m`	`θ < 5 mrad, 0.60 ≤ R ≤ 0.95; inner edge optics-dependent`	AC-LGAD, in vacuum	geometry file (current main)
ZDC	`z ≈ 37.5 m`	`θ < 4 mrad, neutrals`	W/Si + crystal	design value

Table 7 — the far-forward systems. The pot and off-momentum positions are read from `compact/far_forward/roman_pots_eRD24_design.xml` and `offM_tracker.xml` on the current main branch [4] and are not the 26 / 28 and 22.5 / 24.5 m the Yellow Report, the ePIC wiki and several papers still quote; only the angular windows enter the routing.

The pot aperture, measured. An intact ^{6}Li shot through the ePIC far-forward geometry (`tools/fullsim`, September-2024 `epic-main`, 84 points in p_T × azimuth, one event per point, no beam envelope) clears the pots at $|\theta_x| \gtrsim 2.00$ / 1.35 / 1.03 mrad in the 5 × 41 / 10 × 100 / 18 × 275 optics against $|\theta_y| \gtrsim 3.0$ / 3.0 / 2.3 mrad: the aperture opens horizontally, because the beamline images an IP angle onto the pot plane with a horizontal lever R_{12} = 30.6 m (measured at 18 × 275) against a vertical few metres. The geometry has since moved (Report 4 §6): 16 mm modules replace 32 mm ones, the insertion carries per-energy 10σ offsets, and the 1 mm aluminium RF shields are commented out — in the direction of a smaller blind block at the top energy and a larger one at 5 × 41. Every aperture-conditional number is conditioned on the September-2024 measurement until it is repeated. Status: **measured in a superseded geometry**.

Sensor and readout. AC-LGAD at 500 × 500 μm pitch with a 100 × 100 μm metal electrode over a 30 μm active thickness, bump-bonded to EICROC; timing ≈ 35 ps per plane is a requirement, HPK AC-LGADs measure 20–35 ps in test beam; spatial resolution ≈ 140 μm from the pitch alone, ≈ 20 μm shown with charge sharing, ePIC's stated pot baseline binary 500 μm without it; EICROC provides a 10-bit 25 ps time-of-arrival TDC and an 8-bit 40 MHz SAR ADC for charge [5]. The insertion is quantised by 16 × 16 mm modules with a 2.4 cm base plus a per-energy offset (0.27 / 0.71 / 2.96 cm at 18 × 275 / 10 × 100 / 5 × 41), with the geometry file's own comment that these "are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have" [4]. Rates: a few Hz per channel in normal running, 30–50 kHz at ≈ 10σ from beam halo on STAR experience [7]; in-vacuum cooling ≈ 100 W per sensor layer; no published radiation fluence or dose.

6.2 · Central detector

QUANTITY	VALUE USED	SOURCE	STATUS
EMCal σ_E/E, backward (η < −2)	2%/√E ⊕ 1%	YR Fig. 8.3; the $PbWO_4$ crystal expectation	requirement, correctly sourced

… backward transition ($-2 < \eta < -1$)	`7%/√E ⊕ 1%`	YR Fig. 8.3	stochastic exact; constant 2× optimistic against YR Table 3.1
… barrel ($\|\eta\| < 1$)	`10%/√E ⊕ 1%`	YR Table 8.20 band (10–12)%/√E ⊕ (1–3)%	optimistic corner; constant 2–3× optimistic against the ePIC BIC requirement
… forward ($\eta > 1$)	`7%/√E ⊕ 1%`	—	**mis-sourced**: 7%/√E belongs to $-2 < \eta < -1$; latent, no sweet-spot electron is forward
… noise term	`absent`	YR Table 3.1 has 1–2%/E	missing; 2% at 1 GeV
tracking σ_p/p	`√((a p)² + b²), η-piecewise`	YR Fig. 8.3 / Table 11.2; ATHENA Table 4 quotes the same as "Requirements"	requirement — barrel and forward exact, endcap constants padded; ePIC full simulation reaches 0.45–0.6% at 1 GeV/c, so realistic to ≈ ±20%
tracking σ_θ	`5 / 3 / 2 / 1 / 2 / 3 / 5 mrad by η`	—	**placeholder** — no angular requirement in either YR table; also the model's only stand-in for the 81–211 μrad beam-divergence floor on θ′, so the honest bracket is a factor ≲ 3 (Report 2 §3)
electron identification	`0.70–0.95 by η`	constructed between ATHENA JINST 17 P10019 Table 5 and ECCE NIM A 1055 168464 §3.5.2	anchored at one point; the sources differ by ≈ 25 points in the barrel; cancels in the spin-state ratio
hadron-side coverage, thresholds, efficiencies, noise	`tracks |η| ≤ 3.5,` p_T `> 0.2 GeV, 95%; calorimeters |η| ≤ 3.7; 50 MeV noise`	—	stand-ins of the right magnitude (Report 2 Table 2); the ePIC nominal forward reach is 4.0
hadron-side resolutions	`YR tables per region, 2% EMCal constant`	YR requirement tables	requirement

Table 8 — the central-detector model, fact-checked against the Yellow Report. Every sweet spot of the programme puts the scattered electron backward ($\eta = -2.9$ to -1.6), where the calorimeter table is right to within 20% and where the tracker is the better measurement anyway; the mis-sourced forward row is latent. The angular resolution is the entry to watch: the 0.1 mrad quoted in some ePIC talks is a smearing input, not a measured resolution, and would delete the beam-divergence floor the placeholder stands in for.

7 · What the programme assumes that the specification does not give it

#	WHAT IS MISSING	WHAT WE DO INSTEAD	OWNER
1	lithium beam divergence and Δp/p, per configuration and cooling scenario	derive from the proton tables with the γ-matched species step (§4.1)	C-AD / ePIC FF WG
2	a lithium tagging optics: can IR6 be matched with the horizontal hadron β* raised 50–176× — the electron β* co-de-squeezed so ξ_p stays ≤ 0.015 — and what are the IP-to-pot phase advances (parallel-to-point transport)?	assume the optimum of §4.2 exists; without it no ^{6}Li far-forward tag exists at IP6 (§5)	C-AD
3	lithium emittance, intensity, β* and luminosity — nothing published	assume proton-class ε_N (gold calibration, Z^3/A^2 IBS scaling) and the Yellow Report per-nucleon luminosity	C-AD / ANL source group
4	what physically sets the 10σ retraction — collimation hierarchy, halo lifetime, or convention — and pots that follow the envelope of a de-squeezed optics	assume 10σ and pots that move (§4.2, Report 4 §3)	C-AD / ePIC FF WG
5	EICROC input charge dynamic range in fC, and LGAD gain suppression at ≈ 9 MIP	assume the charge is usable; the far-forward Z-identification depends on it entirely (Report 4 §4)	OMEGA-IJCLab / BNL AC-LGAD group
6	per-optics transfer matrix at the pot planes — R_{11}, R_{12}, R_{21}, R_{22}, D, D′	use R_{12} = 30.6 m, measured at 18 × 275 only, and work in angle elsewhere	ePIC FF WG

7	ePIC tracking angular resolution and electron identification — no published curves	η-piecewise stand-ins, flagged in the code and in Table 8	ePIC
8	the calorimeter noise and threshold floor on the hadronic $E - p_z$ sum at Σ_h = 0.2–0.5 GeV	50 MeV Gaussian stand-in; it sets the y ≈ 0.01 bins (Report 2 §4.2)	ePIC inclusive WG
9	radiation environment at the Roman Pots — no published fluence or dose	not modelled	ePIC FF WG
10	polarization-preservation scheme for ^{6}Li, which cannot use Siberian snakes	assume it works at the EPIOS source targets	C-AD / EPIOS
11	no EIC document specifies a tensor-polarization scale uncertainty, for any species; the 1% requirement is vector-only	ours to specify: $\delta P_{zz}/P_{zz} \le 5\%$ (Report 2 §6), a 1:1 normalisation on Δ	ANL source + BNL polarimetry
12	relative luminosity between fills of different P_{zz} is unspecified; the $R < 10^{-4}$ requirement is bunch-by-bunch within a fill	ours to specify: $\le 10^{-3}$ on the ratio, or bunch-by-bunch alternation of the tensor states, which also makes the acceptance cancellation exact	EIC project / ePIC luminosity / source
13	EICrecon has no light-ion far-forward configuration: the transfer matrix is selected by matching the beam proton PDG code	this repository does its own transport; the deliverable is a beamline configuration and a transfer matrix EICrecon can adopt	this programme
14	Roman-Pot commissioning is deferred past year 1 and conditional on beam conditions	projections assume the pots exist from the start	ePIC
15	a 9 GeV electron beam is the early-science baseline; no ESR lattice, emittance, β* or polarization table exists for it	only 5, 10 and 18 GeV are modelled	C-AD

Table 9 — the assumptions, ranked by leverage on a published number. Items 1–10 and 14–15 are machine and detector inputs, to be asked for precisely; 11 and 12 are properties of a measurement nobody has designed yet and are this programme's to specify; 13 is the point of the development rather than a gap in the specification.

8 · Summary

The electron and proton beams are specified in detail and this programme uses them as specified; the gold rows calibrate the one derivation that matters, the lithium divergence; the ion energies follow from the revolution-period constraint and are checked against the published gold and ^{3}He menus. For lithium itself essentially nothing exists beyond a top energy and a G factor, so §4 is a derivation and not a citation, and its consequence is the sharpest statement this report makes: at every published optics the intact ^{6}Li recoil and the ^{6}Li α spectator are inside the beam envelope, and the far-forward half of the polarized-lithium programme exists only with a tagging optics — a one-plane β* de-squeeze of 50–176 with pots that follow it, at 1/7–1/13 of the luminosity — which is the request to put to C-AD. The detector is specified where it matters most for the tags, in a geometry file that can be read, and least where the central reconstruction lives, which is why Table 8 marks the track angular resolution a placeholder and the electron-identification curve a construction between two non-ePIC sources, rather than giving numbers that would look like a specification.

References

[1] R. Abdul Khalek et al., *Science requirements and detector concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (`refs/2103.05419_part1.pdf` … `refs/2103.05419_part4.pdf`): Tables 10.1 and 10.2 (beam parameters per configuration and optics), Table 11.48 (Roman-Pot p_T smearing), §11.6 (far forward), §14.5.3 (superconducting nanowires), and the detector requirement tables.

[2] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794 (`refs/2510.10794.pdf`): Table I (HERA versus EIC parameters); the species table with G and mc²/G; the synchronisation mechanism (pp. 12–13); the polarized-lithium case.

[3] E. Hamwi, G. H. Hoffstaetter, *Polarization transmission in the Electron-Ion Collider's Hadron Storage Ring*, Phys. Rev. Accel. Beams 29 (2026) 073501, arXiv:2509.18558 (`refs/2509.18558.pdf`): Table I, the species-dependent resonance spectrum; small-G ions cannot use Siberian snakes.

[4] eic/epic, branch main: `compact/far_forward/roman_pots_eRD24_design.xml` , `offM_tracker.xml` and `compact/fields/beamline_*.xml` , read directly 2026-08-26/27: station positions, module size, per-energy insertions, the RF-

shield comment.

[5] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, DIS 2023 (`refs/ePIC_far_forward_talk_DIS_2023_v2.pdf`), and the ePIC Collaboration Meeting, JLab, July 2025: acceptances, the AC-LGAD specification, the in-vacuum cooling load, the x-moving layer.

[6] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (`refs/2108.08314.pdf`): Table I, the far-forward acceptance windows the router uses.

[7] ePIC Collaboration, *Preliminary Design Report*, September 2024: Roman-Pot rates, a few Hz per channel in normal running and 30–50 kHz at ≈ 10σ from beam halo.

[8] This repository: Report 1 §6.1 and `evgen/scripts/tagging_optics.py` (the tagging optics priced), Report 2 (the reconstruction chain), Report 4 (the near-beam study), `plans/10_beam_divergence_light_ions.md` (the derivation record), `fastsim/scripts/tagging_acceptance.py` (Table 6), `plans/05_doubly_polarized_generator.md` §5.2 (the cluster momentum density and its β band), `plans/03_phase2_full_simulation.md` §2.2 and `tools/fullsim/README.md` (the far-forward routing and the particle-gun scan).

Appendix A · Revision history

DATE	REVISION
2026-08-28	Rewritten as a paper. The far-forward optics are now per configuration throughout the code (`farforward.yr_optics`, `tagging_optics`, a rectangular envelope applied to each fragment's azimuth in the spectator routing — at every aspect ratio since a 2026-08-28 review, an isotropic envelope having until then fallen back on its inscribed circle), and §5 with Table 6 is new: the lithium tags at the Yellow Report optics and at the tagging optics of Report 1 §6.1, replacing the single proton-derived 73 µrad the fast simulation had applied at every configuration, which gave the ^{6}Li α tag as 1.85% at 18 × 137.5 and 20% at 5 × 40.8 against 1.6% and 1.7% here. §4.2 states the tagging optics as a proposal; §7 adds the two questions it raises for C-AD and the calorimeter noise floor. β = 0.30 GeV is defined where it is first used, and Table 6's triton row is qualified: the routing carries no over-rigid branch, and the particle-gun scan of `tools/fullsim` contradicts "no coverage" without settling it. §5 gained the ^{7}Li luminosity-cost paragraph — the tagging optics is a factor 8–15 net loss for ^{7}Li, so the two isotopes are separate runs — and Table 6's caption the two-sampler note: S wave here, S + D in Report 4 §2.1, 1.5–1.7% against 2.5–2.9% (plans/09 B2, B3). The ^{7}Li residual between the two samplers is a difference of acceptance definition and not of density, and like for like they agree to 0.1 point: the pure model's Roman-Pot tag is 0.9568 / 0.9668 / 0.9721, and inside the k ≤ 1.2 GeV/c on which the tagged model's momentum grid ends 0.9626 / 0.9683 / 0.9736, against the tagged sampler's 0.9614 / 0.9676 / 0.9726; at 10 × 99.5 and 18 × 117.9 the B0 fraction is zero and the two definitions coincide. With $P_D = 0$ the two ^{6}Li samplers agree quantile by quantile.
2026-08-27	First version, after the γ-matching correction (plans/10): the ion energy menu, the beam tables, the lithium divergence derived, the far-forward geometry read from the current main branch, the central-detector table fact-checked against the Yellow Report (the tracking resolution is the YR requirement and not a placeholder; the calorimeter forward row is mis-sourced; the noise term is missing), and the list of unknowns with owners. Later the same day: the synchronisation mechanism verbatim from EPIOS and the conflict with Table 10.1's 100 GeV row; store cooling not in the baseline.

Status vocabulary: **design value** — appears in a design document as a parameter; **requirement** — a specification the subsystem must meet; **measured** — test beam, operations, or a simulation of the geometry; **derived** — computed here from published inputs; **unknown** — not published anywhere we could reach. Beam divergences and Δp/p are Yellow Report Tables 10.1 and 10.2 unless stated; ion energies are computed by `fastsim/polli_fastsim/beams.py` and pinned by `fastsim/tests/test_beams.py`; the far-forward optics by `polli_fastsim/farforward.py` and `fastsim/tests/test_optics_20260828.py`. Built by `reports/build_report.py`.